

MH1100 Calculus I

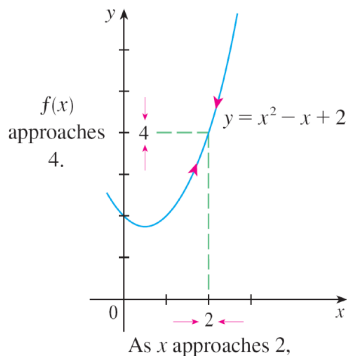
Handout 4 - Limits III

The precise definition of a limit

- The precise definition of a limit
- The precise definitions of one-sided limits
- Using estimation to prove complicated limits
- Using the triangle inequality to prove the Sum Law
- The precise definitions of infinite limits

The precise definition of a limit

Recall the idea of **limit**



$\lim_{x \rightarrow 2} f(x) = 4 \Leftrightarrow$ As x get closer and closer to 2,
 $f(x)$ gets closer and closer to 4.

The precise definition of a limit

Recall the intuitive definition of a limit.

Intuitive Definition of a Limit

Let $f(x)$ be a function defined on some open interval that contains a , except possibly at a itself. Then we say that **the limit of $f(x)$ as x approaches a is L** , and we write

$$\lim_{x \rightarrow a} f(x) = L$$

if we can ^{the goal} make all the values of $f(x)$ as close to L as we want, by restricting the variable x to points close enough to a but excluding a itself.
^{the how}

To obtain the precise mathematical definition by simply replacing each blue (vague) part with a mathematical equation that says the same thing.

The precise definition of a limit

Precise Definition of a Limit

Let $f(x)$ be a function defined on some open interval that contains a , except possibly at a itself. Then we say that **the limit of $f(x)$ as x approaches a is L** , and we write

$$y = y(x) : y = 1/x, y = x^2$$

$$\lim_{x \rightarrow a} f(x) = L$$

$\delta = \delta(\epsilon)$ $\Rightarrow$ if ϵ is given, we can find δ
(δ is a function of ϵ)

if every number $\epsilon > 0$ there is a number $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta \text{ then } |f(x) - L| < \epsilon$$

x must be within a distance
of δ from a
but $x \neq a$

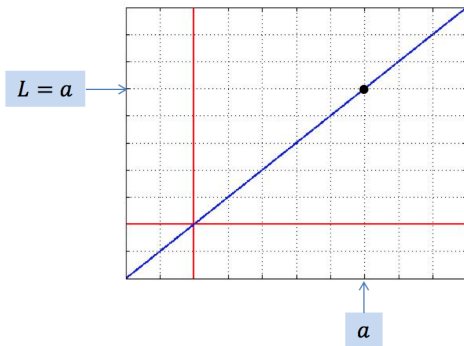
f have a distance less than ϵ from L

The precise definition of a limit

Question.

How would we actually use this ϵ - δ technique to prove some fact about limits? For example, what about the basic limit law

$$\lim_{x \rightarrow a} x = a?$$



Use the ϵ - δ technique to prove $\lim_{x \rightarrow a} x = a$

Answer.

To prove a particular limit our job is to explain why we can always choose an appropriate $\delta > 0$, no matter what $\epsilon > 0$ is.

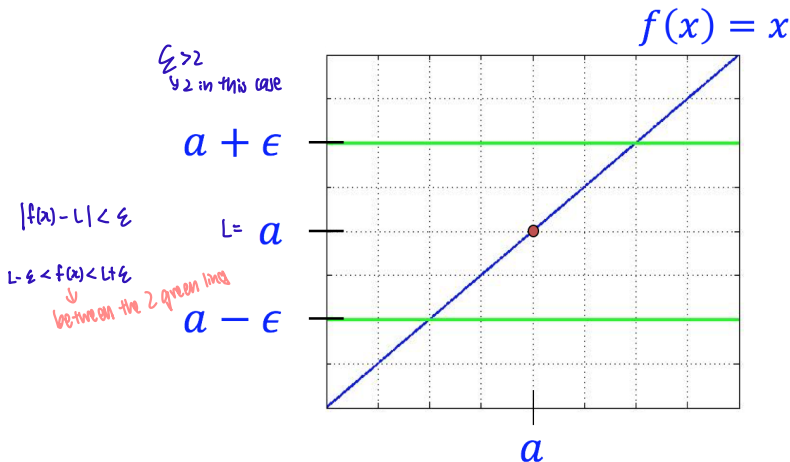
We usually do this in two steps:

infinite choice for $\delta = \delta(\epsilon)$

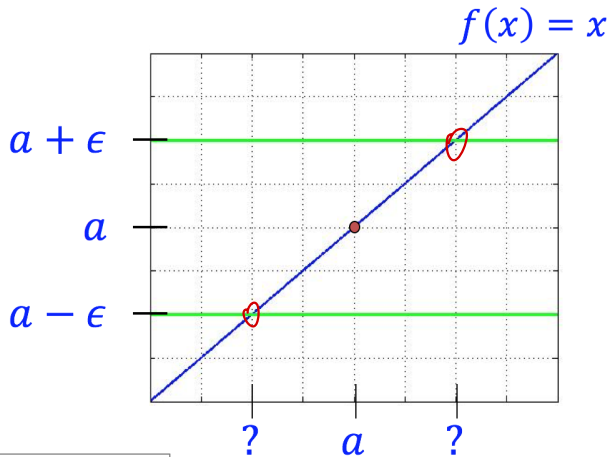
- To begin, we need to explain what δ we will choose when we are given some ϵ . Formally speaking, we need to find a good function $\delta(\epsilon) : \mathbb{R}^+ \rightarrow \mathbb{R}^+$. (Here $\mathbb{R}^+$ denotes $\{x \in \mathbb{R} | x > 0\}$.)
- Then we must show using equations that this choice of $\delta(\epsilon)$ works, no matter what ϵ is.

Use the ϵ - δ technique to prove $\lim_{x \rightarrow a} x = a$

What should we choose for the function $\delta(\epsilon)$?



Use the ϵ - δ technique to prove $\lim_{x \rightarrow a} x = a$

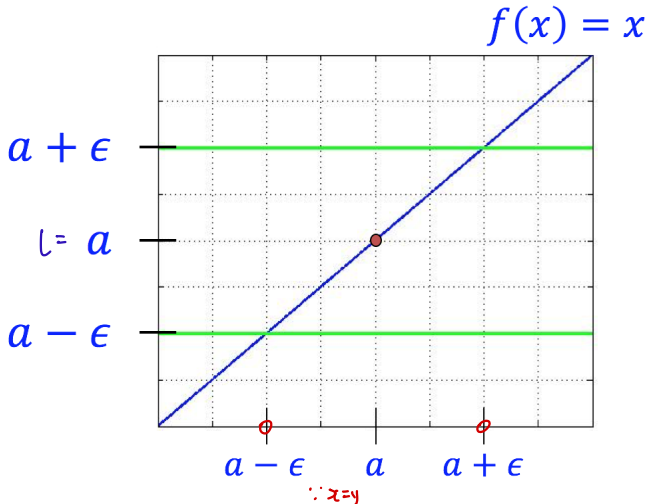


To decide what $\delta(\epsilon)$ should be, we should find these numbers first.

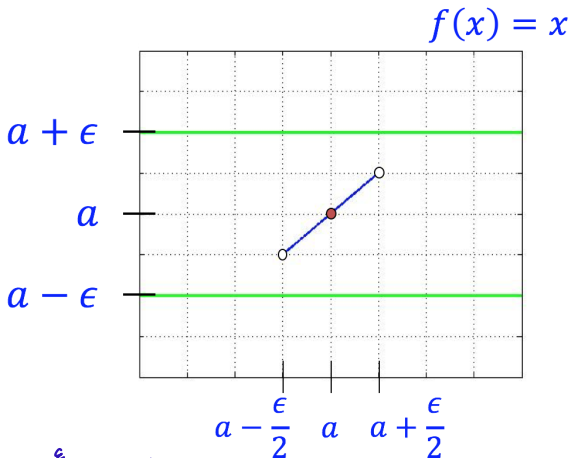
Use the ϵ - δ technique to prove $\lim_{x \rightarrow a} x = a$

Solving $f(x) = a - \epsilon$ gives $x = a - \epsilon$ for the left-most point. Similarly, we solve $f(x) = a + \epsilon$ to get $x = a + \epsilon$ for the right-most point.

$$\epsilon > 0$$
$$|f(x) - l| < \epsilon$$



Use the ϵ - δ technique to prove $\lim_{x \rightarrow a} x = a$



$$0 < |x - a| < \delta$$

$$\delta = \delta(\epsilon) = \frac{\epsilon}{8} \text{ also works!}$$

**just need to show one working δ .*

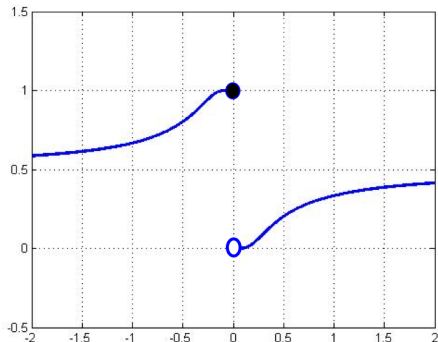
Setting $\delta(\epsilon) = \epsilon/2$ would work, and setting $\delta(\epsilon) = \epsilon$ would work just as well.

In fact, you can set $\delta(\epsilon)$ to anything at all, as long as $\delta(\epsilon) \leq \epsilon$ for all $\epsilon > 0$.

Use the ϵ - δ technique to show that the limit does not exist

Another question.

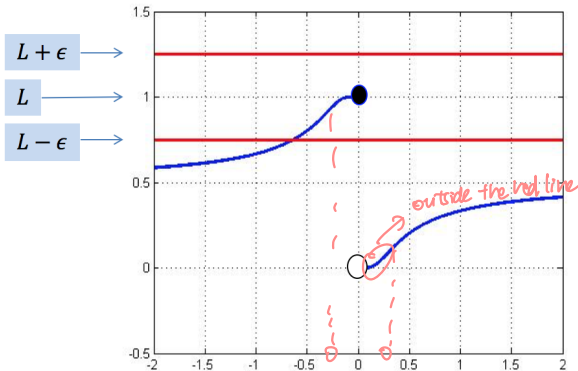
Consider the following function $f(x)$. How would we use the ϵ - δ technique to show that the limit $\lim_{x \rightarrow 0} f(x)$ does not exist?



Use the ϵ - δ technique to show that the limit does not exist

Just say if $\epsilon = \frac{1}{4}$, can you give a working δ ?

$$|f(x) - L| < \epsilon$$
$$L = 1$$



*Will be the same case even if we think $L=0$.

No. You cannot choose a $\delta > 0$ small enough to make the graph on the interval $(a - \delta, a + \delta)$ except possibly at a entirely within $(L - \epsilon, L + \epsilon)$.
 with always have a segment outside the 2 red lines

The precise definition of a limit

We can also reformulate the precise definition in terms of intervals by observing that the inequality $|x - a| < \delta$ is equivalent to $-\delta < x - a < \delta$, which in turn can be written as $a - \delta < x < a + \delta$.

Also $0 < |x - a|$ is true if and only if $x - a \neq 0$, that is, $x \neq a$.

Similarly, the inequality $|f(x) - L| < \epsilon$ is equivalent to the pair of inequalities $L - \epsilon < f(x) < L + \epsilon$.

Therefore, in terms of intervals, the precise definition can be stated as follows:

$$\lim_{x \rightarrow a} f(x) = L$$

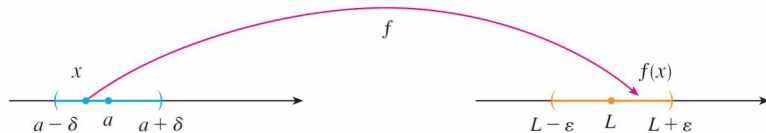
means that for every $\epsilon > 0$ (no matter how small ϵ is) we can find $\delta > 0$ such that if x lies in the open interval $(a - \delta, a + \delta)$ and $x \neq a$, then $f(x)$ lies in the open interval $(L - \epsilon, L + \epsilon)$.

$$\Downarrow \\ |f(x) - L| < \epsilon$$

$$\Downarrow \\ 0 < |x - a| < \delta$$

The precise definition of a limit

The definition of limit says that if any small interval $(L - \epsilon, L + \epsilon)$ is given around L , then we can find an interval $(a - \delta, a + \delta)$ around a such that f maps all the points in $(a - \delta, a + \delta)$ (except possibly a) into the interval $(L - \epsilon, L + \epsilon)$.



The precise definition of a limit

Example 2

Prove that $\lim_{x \rightarrow 3} (4x - 5) = 7$.

Solution.

1. Preliminary analysis of the problem (guessing a value for δ).

Let ϵ be a given **positive number**. We want to find a number δ such that $\delta > 0$

$$\text{if } 0 < |x - 3| < \delta \text{ then } |(4x - 5) - 7| < \epsilon$$

δ unknown ϵ known

But

$$|(4x - 5) - 7| = |4x - 12| = |4(x - 3)| = 4|x - 3|$$

Therefore we want to find δ such that

$$\text{if } 0 < |x - 3| < \delta \text{ then } 4|x - 3| < \epsilon$$

that is,

$$\text{if } 0 < |x - 3| < \delta \text{ then } |x - 3| < \frac{\epsilon}{4}$$

To make sure that $|x - 3| < \frac{\epsilon}{4}$, we should choose $\delta = \epsilon/4$.

Example 2—Solution (Cont'd)

2. Proof (showing that this δ works).

Given $\epsilon > 0$, choose $\delta = \epsilon/4$. If $0 < |x - 3| < \frac{\epsilon}{4}$, then

$$|(4x - 5) - 7| = |4x - 12| = |4(x - 3)| = 4|x - 3| < 4\delta = 4\left(\frac{\epsilon}{4}\right) = \epsilon$$

Thus

$$\text{if } 0 < |x - 3| < \delta \quad \text{then} \quad |(4x - 5) - 7| < \epsilon$$

Therefore, by the definition of a limit,

$$\lim_{x \rightarrow 3} (4x - 5) = 7.$$

Comment: In the solution there were two stages—guessing and proving. We made a preliminary analysis that enabled us to guess a value for δ . But then in the second stage we had to go back and prove in a careful, logical fashion that we had made a correct guess. This procedure is typical of much of mathematics.

The precise definition of a limit

How to find algebraically a δ for a given f , L , a , and $\epsilon > 0$

The process of finding $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \text{whenever} \quad 0 < |x - a| < \delta$$

can be accomplished in two steps.

- Solve the inequality $|f(x) - L| < \epsilon$ to find an open interval (c, d) containing a on which the inequality holds for all $x \neq a$. Note that we do not require the inequality to hold at $x = a$. It may hold there or it may not, but the value of f at $x = a$ does not influence the existence of a limit.
- Find a value of $\delta > 0$ that places the open interval $(a - \delta, a + \delta)$ centered at a inside the interval (c, d) . The inequality $|f(x) - L| < \epsilon$ will hold for all $x \neq a$ in this δ -interval.

$(a - \delta, a + \delta) \subset (c, d)$
 is a subset.

The precise definition of a limit

Exercise

Prove that $\lim_{x \rightarrow 2} (1 - \frac{x}{4}) = \frac{1}{2}$.

1. Guessing a value for δ .

Let ϵ be a given positive number. We want to find a number δ such that if $0 < |x - 2| < \delta$ then $|1 - \frac{x}{4} - \frac{1}{2}| < \epsilon$

But

$$|1 - \frac{x}{4} - \frac{1}{2}| = |\frac{1}{2} - \frac{x}{4}| = \frac{1}{4} |2 - x| = \frac{1}{4} |x - 2|$$

Therefore we want δ such that

if $0 < |x - 2| < \delta$ then $\frac{1}{4} |x - 2| < \epsilon$

that is,

if $0 < |x - 2| < \delta$ then $|x - 2| < 4\epsilon$

This suggests that we should choose $\delta = 4\epsilon$. *also can choose $\delta = 2\epsilon$

The precise definitions of one-sided limits

The intuitive definitions of one-sided limits can be precisely reformulated as follows.

Definition of Left-Hand Limit

$$\lim_{x \rightarrow a^-} f(x) = L$$

if for every number $\epsilon > 0$ there is a number $\delta > 0$ such that $\delta = \delta(\epsilon)$

$$\text{if } a - \delta < x < a \text{ then } |f(x) - L| < \epsilon$$

Definition of Right-Hand Limit

$$\lim_{x \rightarrow a^+} f(x) = L$$

if for every number $\epsilon > 0$ there is a number $\delta > 0$ such that

$$\text{if } a < x < a + \delta \text{ then } |f(x) - L| < \epsilon$$

The precise definitions of one-sided limits

Example 3

Prove that $\lim_{x \rightarrow 4^+} \sqrt{x} = 2$.

Solution.

1. Guessing a value for δ .

Let ϵ be a given positive number. We want to find a number δ such that

$$\text{if } 0 < x - 4 < \delta \text{ then } |\sqrt{x} - 2| < \epsilon$$

that is,

$$\text{if } 4 < x < 4 + \delta \text{ then } \sqrt{x} < 2 + \epsilon$$

or, squaring both sides of the inequality $\sqrt{x} < 2 + \epsilon$, we get

$$\text{if } 4 < x < 4 + \delta \text{ then } x < 4 + 4\epsilon + \epsilon^2 \Leftrightarrow |\sqrt{x} - 2| < \epsilon$$

This suggests that we should choose $\delta = \epsilon^2 + 4\epsilon$.

largest possible δ

*also can choose $\delta = 4\epsilon$
 $\delta = 2\epsilon$
 $\delta = \epsilon^2$*

Example 3—Solution (Cont'd)

2. Showing that this δ works.

Given $\epsilon > 0$, choose $\delta = \epsilon^2 + 4\epsilon$. If $0 < x - 4 < \delta = \epsilon^2 + 4\epsilon$, then

$$4 < x < \epsilon^2 + 4\epsilon + 4 = (\epsilon + 2)^2$$

So

$$2 < \sqrt{x} < \epsilon + 2$$

That is

$$0 < \sqrt{x} - 2 < \epsilon,$$

indicating that

$$|\sqrt{x} - 2| < \epsilon.$$

Therefore, by the definition of right-hand limit,

$$\lim_{x \rightarrow 4^+} \sqrt{x} = 2.$$

Example 3—Solution (Cont'd)

Prove $\lim_{x \rightarrow 4} \sqrt{x} = 2$

$$|x-2| < \varepsilon \stackrel{x \rightarrow 4^-}{\Rightarrow} 2 - \sqrt{x} < \varepsilon$$
$$\sqrt{x} > 2 - \varepsilon$$

The precise definition of a limit

Example A

Prove that $\lim_{x \rightarrow 0} x^{2023} [1 + \sin^2(2024x)] = 0$.

Solution.

1. Guessing a value for δ .

Let ϵ be a given positive number. We want to find a number δ such that

$$\text{if } 0 < |x - 0| < \delta \quad \text{then} \quad |x^{2023}[1 + \sin^2(2024x)] - 0| < \epsilon.$$

But

$$|x^{2023}[1 + \sin^2(2024x)] - 0| = |x^{2023}[1 + \sin^2(2024x)]| = |x^{2023}| \cdot |1 + \sin^2(2024x)|$$

$0 < \sin^2(2024x) < 1$
 $\therefore 1 < 1 + \sin^2(2024x) < 2$

$$\leq 2|x^{2023}| < \epsilon$$

If $|f(x) - L| < g(x)$
 $g(x) < \epsilon$
then $|f(x) - L| < \epsilon$

Question. How can we process from $|x^{2023}| \cdot |1 + \sin^2(2024x)|$ to $|x - 0|$?

Answer. We can replace the complicated term $|1 + \sin^2(2024x)|$ with a much simpler expression 2. This replacement will help because $|1 + \sin^2(2024x)| \leq 2$.

Example A—Solution (Cont'd)

Then, using $0 < |x - 0| < \delta$, we have

$$\begin{aligned} |x^{2023}[1 + \sin^2(2024x)] - 0| &= |x^{2023}| \cdot |1 + \sin^2(2024x)| \\ &\leq 2|x^{2023}| < 2\delta^{2023} = \epsilon \end{aligned}$$

This suggests that we should choose $\delta = \sqrt[2023]{\epsilon/2}$. $2|x^{2023}| > 1 \text{ for } |x| > 1$

2. Showing that this δ works.

Given $\epsilon > 0$, choose $\delta = \sqrt[2023]{\epsilon/2}$. If $0 < |x - 0| < \delta = \sqrt[2023]{\epsilon/2}$, then

$$\begin{aligned} |x^{2023}[1 + \sin^2(2024x)] - 0| &= |x^{2023}[1 + \sin^2(2024x)]| \\ &\leq 2|x^{2023}| < 2(\epsilon/2) = \epsilon \end{aligned}$$

Therefore,

$$\lim_{x \rightarrow 0} x^{2023} [1 + \sin^2(2024x)] = 0.$$

Prove complicated limits using the estimation trick

Just as shown in Example A, it is not always easy to prove that limit statements are true using the ϵ, δ definition.

Sometimes we can prove complicated limits using a trick called **estimation**.

The main idea of estimation

Estimation is a useful trick which can sometimes help us prove complicated limits.

It basically means replacing a *complicated* piece of some expression by something larger, but easier to understand.

Actually, this trick - replacing complicated functions by simpler, larger functions - is useful in many different parts of mathematics, not just calculus.

If $0 < |x-a| < \delta$, then $|g(x)-L| < \epsilon$ } $\Rightarrow$ if $0 < |x-a| < \delta$, $|f(x)-L| < \epsilon$
 $|f(x)-L| < g(x)$ *The task is to find a new function g .

Prove complicated limits using the estimation trick

Example 4

Use estimation to prove that $\lim_{x \rightarrow 3} x^3 = 27$.

Solution.

1. Guessing a value for δ .

Let ϵ be a given positive number. We want to find a number $\delta > 0$ such that

$$\text{if } 0 < |x - 3| < \delta \text{ then } |x^3 - 27| < \epsilon.$$

To connect $|x^3 - 27|$ with $|x - 3|$, we write

$$|x^3 - 27| = |(x - 3)(x^2 + 3x + 9)| = |x - 3| \cdot |x^2 + 3x + 9|.$$

Then we want

$$\text{if } 0 < |x - 3| < \delta \text{ then } |x - 3| \cdot |x^2 + 3x + 9| < \epsilon.$$

Question. How can we process from $|x - 3| \cdot |x^2 + 3x + 9|$ to $|x - 3|$?

Answer. We can try to replace the complicated term $|x^2 + 3x + 9|$ with a simpler positive constant C such that $|x^2 + 3x + 9| < C$ around the point $x = 3$.

$$\begin{aligned} a^3 - b^3 &= (a - b)(a^2 + ab + b^2) \\ a^3 + b^3 &= (a + b)(a^2 - ab + b^2) \end{aligned}$$

restrict into this range

$$(3 - \delta, 3) \cup (3, 3 + \delta) \subset (3 - \delta, 3 + \delta)$$

place a constant on 2

$$\begin{aligned} &\downarrow \\ |x - 3| &< \frac{\epsilon}{C} \end{aligned}$$

Example 4—Solution (Cont'd)

Two constraints
 $\textcircled{1} x \in (3-d, 3+d) \Rightarrow |x^2+3x+9| < C$
 $\textcircled{2} |x-3| < \frac{\epsilon}{C} \Rightarrow |x^2+3x+9| < \epsilon$
 $\Rightarrow |x^2-27| < \epsilon$
 $x \in (3-\delta, 3+\delta) \Rightarrow |x^2-27| < \epsilon$
 $(3-d, 3+d) \cap (3-\frac{\epsilon}{C}, 3+\frac{\epsilon}{C}) = (3-\delta, 3+\delta), \delta = \min(d, \frac{\epsilon}{C})$

If we can find a positive constant C such that $|x^2 + 3x + 9| < C$, then

$$|x^3 - 27| = |(x - 3)(x^2 + 3x + 9)| = |x - 3| \cdot |x^2 + 3x + 9| < C|x - 3|$$

and we can make $C|x - 3| < \epsilon$ by taking $|x - 3| < \epsilon/C$, so we could choose $\delta = \epsilon/C$.

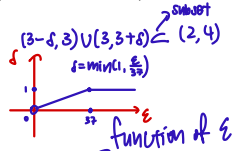
However, $|x^2 + 3x + 9|$ has **no upper bound in its domain $\mathbb{R}$** . We can only find such a number C if we restrict x to lie in some interval centered at 3.

In fact, since we are interested only in values of x that are close to 3, it is reasonable to assume that x is within a distance 1 from 3, that is $|x - 3| < 1$. *Chosen for simplicity.*

Then we have $2 < x < 4$, so $19 < x^2 + 3x + 9 < 37$. Thus $|x^2 + 3x + 9| < 37$, and so $C = 37$ is a suitable choice for the constant.

But now there are two restrictions on $|x - 3|$, namely

$$|x - 3| < 1 \quad \text{and} \quad |x - 3| < \frac{\epsilon}{C} = \frac{\epsilon}{37}.$$



To make sure that both of these inequalities are satisfied, we take δ to be the smaller of the two numbers 1 and $\epsilon/37$. The notation for this is $\delta = \min\{1, \epsilon/37\}$.

Example 4—Solution (Cont'd)

2. Showing that this δ works.

Given $\epsilon > 0$, let $\delta = \min\{1, \epsilon/37\}$. If $0 < |x - 3| < \delta$, then

if x within a dist. of 3, $x \neq 3$

$\min(1, \frac{\epsilon}{37}) \leq 1$

$$|x - 3| < 1 \Rightarrow 2 < x < 4 \Rightarrow |x^2 + 3x + 9| < 37.$$

We also have $|x - 3| < \epsilon/37$, so

increasing function in the interval (2,4)

$$|x^3 - 27| = |x - 3||x^2 + 3x + 9| < \frac{\epsilon}{37} \cdot 37 = \epsilon.$$

Therefore,

$$\lim_{x \rightarrow 3} x^3 = 27.$$

$(a-\delta, a+\delta) \setminus \{a\} \subset (a-\delta, a+\delta) \Rightarrow$ to restrict the upper bound for eqn like x^2+3x+9

The triangle inequality

$$\begin{aligned} &|f(x) - L| < \epsilon \\ \text{if } &① g(x) < \epsilon \\ &② |f(x) - L| < g(x) \\ \text{then } &|f(x) - L| < \epsilon \end{aligned}$$

There is another useful tool for simplifying complicated expressions and proving limits.

Theorem: The triangle inequality

Let $a, b \in \mathbb{R}$. Then

$$|a + b| \leq \underbrace{|a| + |b|}_{g(x)}.$$

The triangle inequality is useful in mathematical analysis for determining the best upper estimate on the sum of two numbers.

Here we can use the triangle inequality to prove the Sum Law.

Theorem (Sum law)

Let f and g be functions and let $a \in \mathbb{R}$. If the limits

$$\lim_{x \rightarrow a} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = M$$

both exist, then $\lim_{x \rightarrow a} (f(x) + g(x))$ exists and

$$\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L + M.$$

Proof of the Sum Law (Cont'd)

Proof of the Sum Law.

1. Guessing a value for δ .

Let $\epsilon > 0$ be given. We want to find a number $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta \quad \text{then} \quad |f(x) + g(x) - (L + M)| < \epsilon.$$

Using the triangle inequality, we can write

$$|f(x) + g(x) - (L + M)| = \underbrace{|(f(x) - L)|}_a + \underbrace{|(g(x) - M)|}_b \leq |f(x) - L| + |g(x) - M|.$$

$|a+b| \leq |a|+|b|$

We make $|f(x) + g(x) - (L + M)|$ less than ϵ by making each of the terms $|f(x) - L|$ and $|g(x) - M|$ less than $\epsilon/2$.

Since $\epsilon/2 > 0$ and $\lim_{x \rightarrow a} f(x) = L$, there must exist a number $\delta_1 > 0$ such that

$$\text{if } 0 < |x - a| < \delta_1 \quad \text{then} \quad |f(x) - L| < \frac{\epsilon}{2}.$$

Handwritten notes: $\lim_{x \rightarrow a} f(x) = L$ (with arrow to the limit statement), "can be $\frac{\epsilon}{4}$ while the one below be $\frac{3\epsilon}{4}$ as long as sum $\leq \epsilon$."

Similarly, since $\epsilon/2 > 0$ and $\lim_{x \rightarrow a} g(x) = M$, there must exist a number $\delta_2 > 0$ such that

$$\text{if } 0 < |x - a| < \delta_2 \quad \text{then} \quad |g(x) - M| < \frac{\epsilon}{2}.$$

We can choose $\delta = \min\{\delta_1, \delta_2\}$, the smaller of the numbers δ_1 and δ_2 .

Handwritten note: $\rightarrow$ so that both inequalities hold simultaneously.

Handwritten note: $\lim_{x \rightarrow a} g(x) = M$ (with arrow to the limit statement)

Proof of the Sum Law (Cont'd)

2. Showing that this δ works.

Notice that

$$\text{if } 0 < |x - a| < \delta \text{ then } |x - a| < \delta_1 \text{ and } |x - a| < \delta_2$$

and so

$$|f(x) - L| < \frac{\epsilon}{2} \text{ and } |g(x) - M| < \frac{\epsilon}{2}$$

Therefore,

$$|f(x) + g(x) - (L + M)| \leq |(f(x) - L)| + |(g(x) - M)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

To summarize,

$$\text{if } 0 < |x - a| < \delta \text{ then } |f(x) + g(x) - (L + M)| < \epsilon$$

Thus, by the definition of a limit,

$$\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L + M.$$

Comment: It is not always easy to prove that limit statements are true using the ϵ, δ definition. Fortunately this is unnecessary, the limits of complicated functions can be found rigorously from the Limit Laws without resorting to the definition directly.

Precise definitions of infinite limits

Infinite limits can also be defined in a precise way.

Precise Definition of an Infinite Limit

Let $f(x)$ be a function defined on some open interval that contains a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = \infty$$

means that for every positive number M there is a positive number δ such that

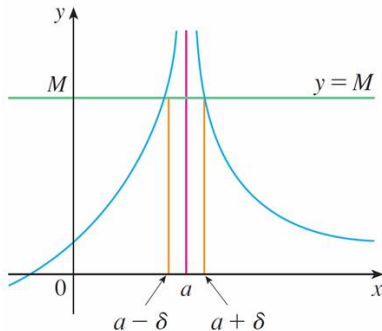
$$\text{if } 0 < |x - a| < \delta \text{ then } f(x) > M$$

This says that the values of $f(x)$ can be made arbitrarily large (larger than any given number M) by requiring x to be close enough to a (within a distance δ , where $\delta(M)$ depends on M , but with $x \neq a$).

value of δ changes with M

Precise definitions of infinite limits

Given any horizontal line $y = M$, we can find a number $\delta > 0$ such that if we restrict x to lie in the interval $(a - \delta, a + \delta)$ but $x \neq a$, then the curve $y = f(x)$ lies above the line $y = M$.



You can see that if a larger M is chosen, then a smaller δ may be required.

Precise definitions of infinite limits

Example 5

Prove that

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty.$$

Solution.

Let M be a given positive number. We want to find a number $\delta > 0$ such that

$$\text{if } 0 < |x - 0| < \delta \text{ then } \frac{1}{x^2} > M.$$

*$h(x) = \sqrt{x}$ is an increasing function
if $x_1 < x_2$, then $h(x_1) < h(x_2)$.
Here $x_1 = x^2$, $x_2 = \frac{1}{M}$*

But

$$\frac{1}{x^2} > M \iff x^2 < \frac{1}{M} \iff \sqrt{x^2} < \sqrt{\frac{1}{M}} \iff |x| < \frac{1}{\sqrt{M}} = \frac{\sqrt{M}}{M}$$

So if we choose $\delta = 1/\sqrt{M}$ and $0 < |x| < \delta = 1/\sqrt{M}$, then $1/x^2 > M$.

This shows that

*largest possible
value for δ .
can choose a
smaller one
like $1/\sqrt{2M}$*

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$$

Precise definitions of infinite limits

Similarly, we can precisely define the negative infinite limit.

Precise definition of a negative infinite limit

Let $f(x)$ be a function defined on some open interval that contains a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = -\infty$$

means that for every negative number N there is a positive number δ such that

$$\text{if } 0 < |x - a| < \delta \quad \text{then } f(x) < N$$

This says that the values of $f(x)$ can be made **arbitrarily small** (smaller than any given number N) by requiring x to be close enough to a (within a distance δ , where $\delta(N)$ depends on N , but with $x \neq a$).

value of f changes when N change.

Precise definitions of infinite limits

Exercise

Prove that $\lim_{x \rightarrow 0^+} \ln x = -\infty$.

Let N be a given number. We want to find a number δ such that if $0 < x < \delta$ then $\ln x < N$.

But $\ln x < N \Rightarrow e^{\ln x} < e^N$
implies $0 < x < e^N$.

Therefore we want δ such that

if $0 < x < \delta$ then $\ln x < N$

that is

if $0 < x < \delta$ then $0 < x < e^N$.

This suggests that we should choose $\delta = e^N$.

$\ln(x) = e^x$ is increasing on $\mathbb{R}$

if $x_1 < x_2$

then $e^{x_1} < e^{x_2}$